

Quiz 2 Formula Sheet

1 Discrete Random Variables and Expectations

1.1 PMFs and Expected Value

For a discrete random variable X , its probability mass function is $p_X(x) = P(X = x)$, where

$$p_X(x) \geq 0 \quad \text{and} \quad \sum_x p_X(x) = 1.$$

The expected value of a function of X is

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x).$$

1.2 Linearity of Expectation

For constants $a_1, \dots, a_n, b$ and random variables $X_1, \dots, X_n$:

$$\mathbb{E}\left[b + \sum_{i=1}^n a_i X_i\right] = b + \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

Linearity of expectation does not require the random variables to be independent.

1.3 Tail Sum Formula

If X takes nonnegative integer values $0, 1, 2, \dots$, then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(X \geq k).$$

2 Indicator Random Variables

2.1 Indicator Expectation

For an event A , define

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases} \quad \text{Then} \quad \mathbb{E}[I_A] = P(A).$$

2.2 Expected Counts

If X counts how many of the events $A_1, \dots, A_n$ occur, then

$$X = \sum_{i=1}^n I_{A_i} \quad \implies \quad \mathbb{E}[X] = \sum_{i=1}^n P(A_i).$$

3 Normal Distribution

For $X \sim N(\mu, \sigma^2)$, the density is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}((x-\mu)/\sigma)^2}, \quad \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = P(Z \leq z) = \int_{-\infty}^z \phi(t) dt.$$

For a standard normal random variable $Z \sim N(0, 1)$, define

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = P(Z \leq z) = \int_{-\infty}^z \phi(t) dt.$$

4 Counting and Standard Discrete Models

4.1 Combinations

For choosing k objects from n distinct objects, and for category counts $x_1 + \cdots + x_r = n$:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{x_1, \dots, x_r} = \frac{n!}{x_1! \cdots x_r!}.$$

4.2 Binomial Distribution

If X counts successes in n independent trials, each with success probability p (and $q = 1 - p$), then

$$X \sim \text{Binomial}(n, p), \quad P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \mathbb{E}[X] = np, \quad \text{SD}(X) = \sqrt{npq}.$$

4.3 Poisson Distribution

If $X \sim \text{Poisson}(\mu)$, then for $k = 0, 1, 2, \dots$,

$$P(X = k) = \frac{e^{-\mu} \mu^k}{k!}, \quad \mathbb{E}[X] = \mu.$$

4.4 Hypergeometric Distribution

Suppose a population has N objects, K of which are successes, and n objects are selected uniformly without replacement. If X is the number of successes selected, then

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad \mathbb{E}[X] = n \frac{K}{N}.$$

4.5 Multivariate Hypergeometric Distribution

Suppose a population is divided into categories of sizes $N_1, \dots, N_r$, where $N_1 + \cdots + N_r = N$. If n objects are selected uniformly without replacement, then for $x_1 + \cdots + x_r = n$:

$$P(X_1 = x_1, \dots, X_r = x_r) = \frac{\prod_{j=1}^r \binom{N_j}{x_j}}{\binom{N}{n}}.$$

4.6 Multinomial Distribution

Suppose n independent trials each result in one of r categories with probabilities $p_1, \dots, p_r$, where $p_1 + \cdots + p_r = 1$. If X_j counts the trials in category j , then for $x_1 + \cdots + x_r = n$:

$$P(X_1 = x_1, \dots, X_r = x_r) = \frac{n!}{x_1! \cdots x_r!} \prod_{j=1}^r p_j^{x_j}.$$



Quiz 2 Review

Exercises

- Roll a die 18 times. Let S = # of ones in all 18 rolls, X_1 = # of ones in the first 7 rolls, and X_2 = # of ones in the last 11 rolls. Find the probability of each event.
 - There are 3 ones in the first 7 trials given there are 5 ones. *Hypergeom ($N=18, K=5, n=7, k=3$)*
 - There are 3 ones in the first 11 trials given that the first six occurred on trial 12. *Binomial ($n=11, p=\frac{1}{6}, k=3$)*
 $\{1, 2, 3, 4, 5\}$
- Roll a 10 sided die 13 times. *Multinomial*
 - Find the probability of this sequence: 111 2222 333333 $(\frac{1}{10})^{13}$
 - Find the probability of 3 ones, 4 twos, 6 threes. (The sequence can be in any order.) $\frac{13!}{3!4!6!} (\frac{1}{10})^{13}$
 - Find the probability of a 3 of a kind, a 4 of a kind, and a 6 of a kind. $10 \cdot 9 \cdot 8 \cdot \frac{13!}{3!4!6!} (\frac{1}{10})^{13}$
 - Find the probability of 3 singles, 2 sets of triples, and a 4 of a kind $(\frac{10}{3})(\frac{7}{2})(\frac{5}{1}) \cdot \frac{13!}{1!1!1!3!3!4!} (\frac{1}{10})^{13}$
- In a poker hand of $n = 5$ cards, find the following probabilities. (Deal 5 cards without replacement.)
 - $P(3 \text{ of a kind, } 2 \text{ of a kind})$ {ranks: 3a, 2b} *13 available ranks, each rank has 4 cards*
 - $P(2 \text{ pairs})$ {ranks: 2a, 2b, c}

order does not matter what a and b is
- Consider a standard deck of cards. Deals cards without replacement.
 - Deal 10 cards. Find the probability of getting 2 sets of triples and 2 sets of pairs.
 - Deal 5 cards to Alice and 5 cards to Bob. Let A be the event that Alice gets a full house (a pair and a triple) and B be the event that Bob gets a full house. Find $P(AB)$.
- A building has $k = 10$ floors above the basement. Let $n = 12$ be the number of people that get into an elevator at the basement. Each person chooses a floor at random to get out, independently of the others. Let X = # of floors that the elevator stops on.

Example: 1, 1, 1, 1, 1, 1, 2, 2, 2, 3, 3, 5 ($x = 4$)

Find:

 - $P(X = 1) = \frac{\binom{10}{1} \cdot 1^{12}}{10^{12}} = \frac{\text{Choose 1 floor} \cdot \text{12 ppl in that floor}}{\text{total outcome}}$
 - $P(X = 2) = \frac{\binom{10}{2} \cdot (2^{12} - 2)}{10^{12}} = \frac{\text{choose 2 floors} \cdot (\text{12 ppl in 2 floors} - \text{all 12 people in either same floor})}{\text{total outcome}}$

Let A_i be the event that the elevator stops on floor i . $= \begin{cases} 1 & \text{if elevator stops on floor } i \\ 0 & \text{otherwise} \end{cases}$

 - $E(X) = \sum_{i=1}^{10} P(A_i) = 10 \times (1 - (\frac{9}{10})^{12})$
 $P(A_i) = P(\text{stops on floor } i) = P(\text{at least one person chooses floor } i) = 1 - P(\text{no one chooses floor } i) = 1 - (\frac{9}{10})^{12}$
- An urn has 20 green out of 50 marbles. The rest are black marbles. Draw marbles without replacement until the 7th green marble. Let X be the number of marbles drawn.

Example: GGGG BBBGGBBG ($x = 12$)

Find:

 - $P(X = x)$ and its support (possible values)
 - $E(X)$ as a simplified fraction

$$= r \cdot \frac{N+1}{K+1} \quad \begin{matrix} r=7 & \text{target green marbles} \\ N=50 & \\ K=20 & \end{matrix}$$

$$= 17$$

Negative hypergeom ($N = 50, K = 20, n = x-1, k = 6$)

$$\downarrow \frac{\binom{20}{6} \binom{30}{x-7}}{\binom{50}{x-1}} \times \frac{20-6}{50-(x-1)}$$

prob of 6 green in first $x-1$ draws Prob draw x is green



7. A p -coin is heads with probability p . Flip a p -coin n times. Let $X = \#$ of repeating sequences of heads of any length. Find $E(X)$.

Example: TTTTTTTT ($n = 8, x = 0$)

Example: HHHH TT HH ($n = 8, x = 2$)

Example: T HHH TT H T ($n = 8, x = 2$)

Find:

- a) $P(X = 0) = P(\text{all are tails}) = (1-p)^n$ HT
THTHTHTH
- b) $P(X = n/2)$ assuming n is even $= P(\text{HTHTHTHT}) + P(\text{THHTHTHT}) = p^{\frac{n}{2}}(1-p)^{\frac{n}{2}} + p^{\frac{n}{2}}(1-p)^{\frac{n}{2}}$
- c) $E(X)$ e.g. $n=8$

Let $I_i = 1$ if a run of Heads starts at flip i , $= 0$ otherwise.

For $i=1$, $E[I_1] = P(\text{Heads}_1) = p$

For $i=2, \dots, n$, $E[I_i] = P(\text{Tails}_{i-1} \cap \text{Heads}_i) = (1-p)p$

$$E[X] = \sum_{i=1}^n E[I_i] = p + \sum_{i=2}^n E[I_i] = p + (n-1)(1-p)p$$

↳ linearity of expectation applies regardless of dependence between adjacency flips

1 Distribution Cheat Sheet

Binomial

Setup: n independent Bernoulli trials, constant p (*with replacement*).

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \mathbb{E}[X] = np \quad SD(X) = \sqrt{np(1-p)}$$

Geometric & Negative Binomial

Setup: *With replacement*, constant p ; sample until the r -th success. X = total # trials (random).

$$\underbrace{P(X = k) = (1-p)^{k-1}p}_{\text{Geometric: } r=1, k=1,2,\dots} \quad \underbrace{P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}}_{\text{Neg. Binomial: general } r, k=r, r+1, \dots}$$

$$\text{Geometric: } \mathbb{E}[X] = \frac{1}{p}, \quad SD(X) = \frac{\sqrt{1-p}}{p} \quad \text{Neg. Binomial: } \mathbb{E}[X] = \frac{r}{p}, \quad SD(X) = \frac{\sqrt{r(1-p)}}{p}$$

Hypergeometric

Setup: Population N , K successes, sample n (*without replacement*).

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}} \quad \mathbb{E}[X] = n \frac{K}{N}$$

Negative Hypergeometric

Setup: Population N , K successes; sample *without replacement* until the r -th success. Total draws X is random; the last draw must be a success.

$$P(X = x) = \frac{\binom{K}{r-1} \binom{N-K}{x-r}}{\binom{N}{x-1}} \cdot \frac{K - (r-1)}{N - (x-1)} \quad \mathbb{E}[X] = r \cdot \frac{N+1}{K+1}$$

Multinomial

Setup: n independent trials, each landing in one of r categories with probabilities $p_1, \dots, p_r$, $\sum p_i = 1$.

$$P(X_1 = x_1, \dots, X_r = x_r) = \frac{n!}{x_1! \dots x_r!} \prod_{j=1}^r p_j^{x_j} \quad \mathbb{E}[X_j] = np_j \quad \text{requires } \sum x_j = n$$

Multivariate Hypergeometric

Setup: Population N split into r categories ($N_1 + \dots + N_r = N$); sample n *without replacement*.

$$P(X_1 = x_1, \dots, X_r = x_r) = \frac{\prod_{j=1}^r \binom{N_j}{x_j}}{\binom{N}{n}} \quad \mathbb{E}[X_j] = n \frac{N_j}{N}$$

Poisson

Setup: Rare events in time/space, or Binomial with large n , small p ($\mu = np$).

$$P(X = k) = \frac{e^{-\mu} \mu^k}{k!} \quad \mathbb{E}[X] = \mu \quad \text{Var}(X) = \mu \quad SD(X) = \sqrt{\mu} \quad (\text{no upper bound on } k)$$

Use when: counting rare, roughly-independent events with no natural “ n ” (calls/hour, typos/page) *or* approximating Binomial when n is large and p is small.

Careful: not a substitute for Hypergeometric (finite population, no replacement) and breaks down if events aren’t independent or the rate isn’t constant.

Normal $N(\mu, \sigma^2)$

Setup: Continuous, symmetric, unimodal; used for limits/approximations.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \mathbb{E}[X] = \mu \quad SD(X) = \sigma \quad Z = \frac{X - \mu}{\sigma}$$

Use when: approximating a sum/average of many independent random variables (CLT), or approximating Binomial when $\sigma = \sqrt{np(1-p)} \geq 3$.

Careful: apply the ± 0.5 continuity correction when approximating a discrete count, and don’t use it when n is small or p is extreme (use Poisson or exact Binomial instead).

2 How to Pick the Right Model

1. Am I replacing what I draw?

→ **With replacement** (or population is effectively infinite) = trials are **independent**, p stays constant.

→ **Without replacement** = trials are **dependent**, the pool shrinks as you draw.

2. Is each draw sorted into 2 buckets, or more than 2?

→ **2 buckets** (success/failure) = “binomial-family” models.

→ $r > 2$ **buckets** (categories) = “multinomial-family” models.

3. Is the number of draws fixed in advance, or random?

→ **Fixed** n , count successes.

→ **Random:** you stop the instant the r -th success appears = “waiting time” models.

	With replacement (independent)	Without replacement (dependent)
2 outcomes, fixed n	Binomial	Hypergeometric
> 2 outcomes, fixed n	Multinomial	Multivariate Hypergeometric
2 outcomes, draw until r-th success	Geometric ($r = 1$) / Negative Binomial	Negative Hypergeometric

Row 1 – counting successes out of a fixed sample (binary outcome)

Binomial(n, p)

Features:

- Replace after each draw $\Rightarrow p$ never changes
- Trials **independent**
- n fixed, chosen *before* sampling
- Only 2 outcomes per trial

Template: “Flip a coin n times”; “roll a die n times, count how many land on 6”; “sample n people with replacement.”

Hypergeometric(N, K, n)

Features:

- No replacement $\Rightarrow$ pool of successes shrinks
- Trials **dependent**
- n fixed, population size N is finite & known
- Only 2 outcomes per trial (success/failure)

Template: “An urn has N balls, K green”; “deal n cards from a 52-card deck”; “sample n items without replacement.”

Worked contrast

Binomial: A factory line makes items with a 5% independent defect rate. Test 20 items (each defective/OK independently). $X = \#$ defective $\Rightarrow X \sim \text{Bin}(20, 0.05)$.

Hypergeometric: A batch of 20 items truly contains exactly 3 defectives. Pick 5 items out to inspect without putting them back. $X = \#$ defective in the sample $\Rightarrow X \sim \text{HGeom}(N = 20, K = 3, n = 5)$.

Row 2 – counting outcomes across more than 2 categories

Multinomial

Features:

- Replace / infinite population $\Rightarrow$ each p_j stays fixed
- Trials **independent**
- n fixed, $r > 2$ categories, $\sum p_j = 1$
- Generalizes Binomial to r outcomes

Template: “Roll a 6-sided die n times, count each face”; “ n people each independently pick one of r candidates.”

Multivariate Hypergeometric

Features:

- No replacement $\Rightarrow$ each category pool shrinks
- Trials **dependent**
- n fixed, population split into r finite groups $N_1, \dots, N_r$
- Generalizes Hypergeometric to r outcomes

Template: “A deck has 4 suits”; “an urn has red/green/blue balls”; deal or draw n without replacement, count each category.”

Worked contrast

Multinomial: Roll a fair 6-sided die 13 times (each roll independent, with replacement since it’s a die). $(X_1, \dots, X_6) =$ counts of each face $\Rightarrow \text{Multinomial}(13; \frac{1}{6}, \dots, \frac{1}{6})$.

Mult. Hypergeometric: A 52-card deck has 4 suits of 13 cards each. Deal 10 cards *without* replacement. $(X_1, X_2, X_3, X_4) = \#$ of each suit dealt $\Rightarrow \text{Mult. HGeom}(N = 52; N_1 = N_2 = N_3 = N_4 = 13; n = 10)$.

Row 3 – counting draws needed to reach the r -th success (random stopping time)

Geometric / Negative Binomial

Features:

- Replace after each draw $\Rightarrow p$ never changes
- Trials **independent**
- Number of draws X is **random**: stop at the r -th success
- Geometric is the special case $r = 1$

Template: “Flip a coin until the 3rd heads”; “keep sampling with replacement until r successes.”

Negative Hypergeometric

Features:

- No replacement $\Rightarrow$ successes & failures both deplete
- Trials **dependent**
- Number of draws X is **random**: stop at the r -th success
- Last draw is *always* a success (that’s what stops you)

Template: “An urn has N balls, K green; draw *without* replacement until the r -th green.”

Worked contrast

Neg. Binomial: A basketball player makes free throws independently with probability $p = 0.7$ each. Shoot until the 4th make. $X = \#$ shots taken $\Rightarrow \text{Neg. Binomial}(r = 4, p = 0.7)$.

Neg. Hypergeometric: An urn has 50 marbles, 20 green. Draw *without* replacement until the 7th green marble appears. $X = \#$ marbles drawn $\Rightarrow \text{Neg. HGeom}(N = 50, K = 20, r = 7)$.

Poisson and Normal are approximations, not sampling models

Poisson

Use Poisson when...

- Counting events over time/space with **no natural** n (calls per hour, typos per page, decays per second)
- Approximating Binomial: n **large**, p **small**, so $\mu = np$ is moderate (rule of thumb: $n \geq 20$ and $p \leq 0.05$, or $np \leq 10$)
- Events are rare and roughly independent of each other

Don't use Poisson when...

- Sampling is **without replacement** from a small, finite population $\rightarrow$ use Hypergeometric exactly
- p isn't small or n isn't large $\rightarrow$ just use exact Binomial
- Events **cluster** or the rate changes $\rightarrow$ mean $\neq$ variance in reality (overdispersion)
- You need a **hard upper bound** on the count (Poisson has none)

Worked contrast

Exact Binomial (don't approximate): $n = 15$ emails, each independently spam with $p = 0.2$. $X = \#$ spam. Here n is small and p isn't tiny, so compute $X \sim \text{Bin}(15, 0.2)$ exactly.

Poisson, no Binomial underneath: Typos are scattered through a book at a constant rate of 0.5 per page, independently from page to page. $X = \#$ typos on one page $\Rightarrow X \sim \text{Poisson}(0.5)$ directly – there's no fixed “ n trials” here.

Poisson as approximation: A shipment of $n = 10,000$ items each has an independent defect probability $p = 0.0003$. $X = \#$ defective. Exact Binomial is painful to compute by hand; since n is huge and p is tiny, approximate with $X \approx \text{Poisson}(\mu = np = 3)$.

Normal

Use Normal when...

- Approximating a **sum or average** of many independent random variables (Central Limit Theorem)
- Approximating Binomial when $\sigma = \sqrt{np(1-p)} \geq 3$ (bigger σ , better approximation)
- You're comfortable applying the $\pm \frac{1}{2}$ **continuity correction** to go from discrete to continuous

Don't use Normal when...

- n is small or $\sigma < 3$ – the approximation is unreliable
- p is close to 0 or 1 without n being huge $\rightarrow$ distribution is skewed; try Poisson or exact Binomial instead
- You forget the continuity correction when asking for a single exact count
- The quantity isn't a sum/count of independent pieces at all

Worked contrast

Normal approx to Binomial: $n = 200$ voters, each independently prefers Candidate A with $p = 0.5$. $X = \#$ preferring A. $\mu = 100$, $\sigma = \sqrt{50} \approx 7.07 \geq 3$, so

$$P(90 \leq X \leq 110) \approx \Phi\left(\frac{110.5 - 100}{7.07}\right) - \Phi\left(\frac{89.5 - 100}{7.07}\right).$$

When NOT to: $n = 10$, $p = 0.9$. Here $\sigma = \sqrt{10(0.9)(0.1)} \approx 0.95 < 3$ – too skewed and too small for Normal; use exact Binomial instead.

	Poisson approximation	Normal approximation
Best when	n large, p small ($\mu = np$ moderate)	n large, $\sigma = \sqrt{np(1-p)} \geq 3$
Shape it captures	Skewed, discrete, no upper bound	Symmetric, continuous
Common error	Using it without replacement (should be Hypergeometric)	Forgetting the ± 0.5 continuity correction

Summary

Model	Replace?	# categ.	n fixed?	What X counts
Binomial	Yes (indep.)	2	Fixed	# successes in n trials
Hypergeometric	No	2	Fixed	# successes in sample of n
Multinomial	Yes (indep.)	$r > 2$	Fixed	counts in each of r categories
Mult. Hypergeometric	No	$r > 2$	Fixed	counts in each of r categories
Geometric / Neg. Binomial	Yes (indep.)	2	Random	# draws to reach r -th success
Negative Hypergeometric	No	2	Random	# draws to reach r -th success
Poisson	N/A	2	N/A	# rare events in an interval, or Binomial approx. ($\mu = np$)
Normal	N/A	N/A	N/A	continuous approx. to a sum/count (CLT, or Binomial approx.)